

# Functions, Injectivity, Surjectivity, Bijections

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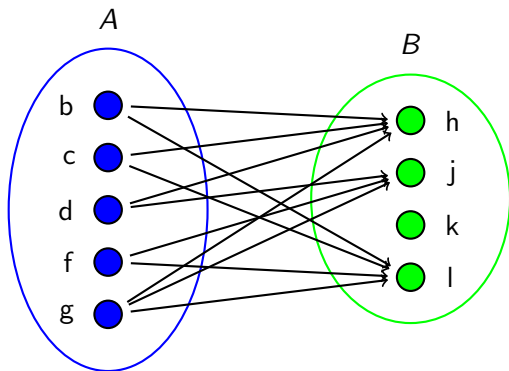
# Overview

Relation Diagrams (4.4.1)

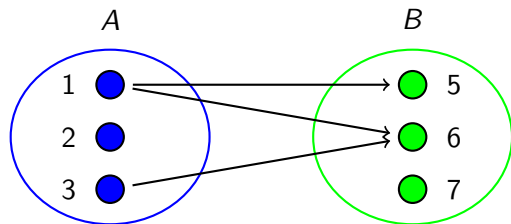
Relational Images (4.4.2)

## Binary relations

Definition. A *binary relation*,  $R$ , consists of a set,  $A$ , called the domain of  $R$ , a set,  $B$ , called the codomain of  $R$ , and a subset of  $A \times B$  called the graph of  $R$ .



# Properties of relations

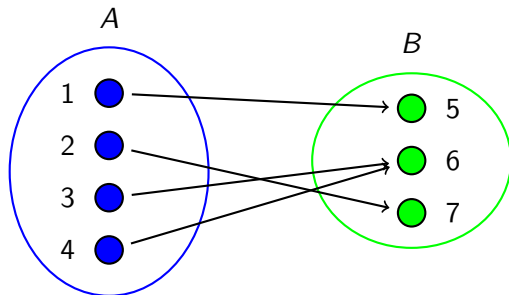


A binary relation:

- ▶ is a *partial function* when it has the  $[\leq 1 \text{ arrow out}]$  property.  
Book: “function”. Us: “function” is  $[= 1 \text{ arrow out}]$  property.
- ▶ is *surjective* when it has the  $[\geq 1 \text{ arrows in}]$  property.
- ▶ is *total* when it has the  $[\geq 1 \text{ arrows out}]$  property.
- ▶ is *injective* when it has the  $[\leq 1 \text{ arrow in}]$  property.
- ▶ is *bijective* when it has both the  $[= 1 \text{ arrow out}]$  and the  $[= 1 \text{ arrow in}]$  properties.

## Example relation #1

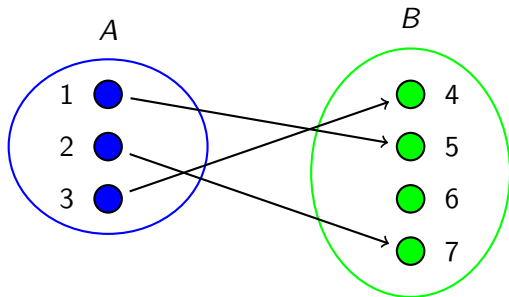
partial function:  $[\leq 1 \text{ out}]$ . surjective:  $[\geq 1 \text{ in}]$ . total:  $[\geq 1 \text{ out}]$ .  
injective:  $[\leq 1 \text{ in}]$ . bijective:  $[= 1 \text{ out}]$  and  $[= 1 \text{ in}]$ .



total, surjective, function.

## Example relation #2

partial function:  $[\leq 1 \text{ out}]$ . surjective:  $[\geq 1 \text{ in}]$ . total:  $[\geq 1 \text{ out}]$ .  
injective:  $[\leq 1 \text{ in}]$ . bijective:  $[= 1 \text{ out}]$  and  $[= 1 \text{ in}]$ .

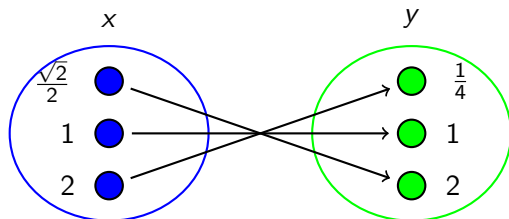


total, injective, function.

## Example relation #3

partial function:  $[\leq 1 \text{ out}]$ . surjective:  $[\geq 1 \text{ in}]$ . total:  $[\geq 1 \text{ out}]$ .  
injective:  $[\leq 1 \text{ in}]$ . bijective:  $[= 1 \text{ out}]$  and  $[= 1 \text{ in}]$ .

Equation  $y = 1/x^2$  on  $\mathbb{R}^+$ .  $x$  is an element in the domain,  $y$  is an element in the co-domain.



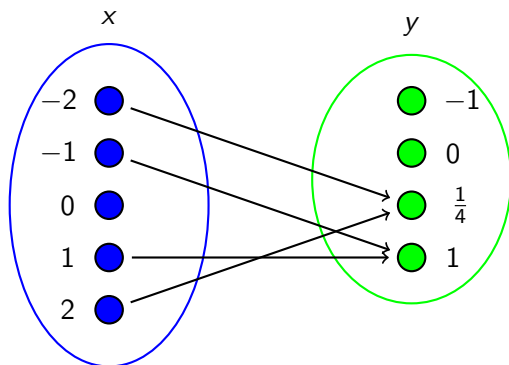
bijective partial function. (implies surjective, total, injective.)

## Example relation #4

partial function:  $[\leq 1 \text{ out}]$ . surjective:  $[\geq 1 \text{ in}]$ . total:  $[\geq 1 \text{ out}]$ .

injective:  $[\leq 1 \text{ in}]$ . bijective:  $[= 1 \text{ out}]$  and  $[= 1 \text{ in}]$ .

Equation  $y = 1/x^2$  on  $\mathbb{R}$ .



partial function.



## Image definition

Definition. The *image* of a set  $Y \subseteq A$  under a relation  $R : A \rightarrow B$ , written  $R(Y)$ , is the subset of elements of the codomain  $B$  of  $R$  that are related to some element in  $Y$ .

In terms of the relation diagram,  $R(Y)$  is the set of points with an arrow coming in that starts from some point in  $Y$ .

$$R(Y) = \{x \in B \mid \exists y \in Y, y R x\}.$$

## Inverse definition

Definition: The *inverse*  $R^{-1}$  of a relation  $R : A \rightarrow B$  is the relation from  $B$  to  $A$  defined by the rule

$b R^{-1} a$  IFF  $a R b$ .

Definition: The image of a set under the relation  $R^{-1}$  is called the *inverse image* of the set. That is, the inverse image of a set  $X$  under the relation  $R$  is defined to be  $R^{-1}(X)$ .

Example:  $x R y$  iff there's a dictionary word with first letter  $x$  and second letter  $y$ . The image  $R(\{c, k\})$  is the letters that can appear after  $c$  or  $k$  at the beginning of a word. It's the set  $\{a, e, h, i, l, n, o, r, u, v, w, y, z\}$ .

The inverse image  $R^{-1}(\{c, k\})$  is the letters that can appear before  $c$  or  $k$  at the beginning of a word. It's the set  $\{a, e, i, o, s, t, u\}$ .

# Inverses of relations

What can we infer about  $R^{-1}$  if  $R$  is:

- ▶ partial function? injective
- ▶ surjective? total
- ▶ total? surjective
- ▶ injective? partial function
- ▶ bijective? bijective
- ▶ function? injective and surjective

## More examples to consider

Make natural examples for each combination of properties.

- ▶  $y = \sqrt{x}$  on  $\mathbb{R}$
- ▶  $y = \sqrt{16 - \sqrt{x}}$  on  $\mathbb{R}$
- ▶  $y = |x + 10|$  on  $\mathbb{Z}$
- ▶  $y = |x \bmod 2|$  on  $\mathbb{Z}$
- ▶  $y = \sin(x)$  on  $\mathbb{R}$
- ▶  $x^2 + y^2 = 10$  on  $\mathbb{R}$